

第5章 贝塞尔函数

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5.1 贝塞尔方程的引出

(Bessel Equation)

引例 设有半径为 **b** 的薄圆盘，其侧面绝热，若圆盘边界上的温度恒保持为零度，且圆盘内初始温度为 $f(x, y)$ ，求圆盘内瞬时温度分布规律.

解 可归结为下述定解问题

$$\begin{cases} \frac{\partial u}{\partial t} = a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), & x^2 + y^2 < b^2, t > 0 \\ u|_{x^2+y^2=b^2} = 0, & t > 0 \\ u|_{t=0} = f(x, y), & x^2 + y^2 < b^2. \end{cases}$$

用分离变量法求解 $u_t = a^2(u_{xx} + u_{yy})$

令 $u(x, y, t) = v(x, y)T(t)$

则 $v(x, y)T'(t) = a^2(v_{xx} + v_{yy})T(t)$

$$\frac{T'(t)}{a^2 T(t)} = \frac{v_{xx} + v_{yy}}{v(x, y)} = -\lambda$$

一次分离

$$u|_{x^2+y^2=b^2} = 0 \Rightarrow v(x, y)|_{x^2+y^2=b^2} = 0$$

边界条件分离

● $T'(t) + \lambda a^2 T(t) = 0 \Rightarrow T(t) = Ke^{-\lambda a^2 t}$

● $\begin{cases} v_{xx} + v_{yy} + \lambda v = 0 \rightarrow \text{二维 Helmholtz 方程} \\ \text{边界条件: } v|_{x^2+y^2=b^2} = 0 \end{cases}$

将亥姆霍兹方程转换为平面极坐标系

$$\begin{cases} \frac{\partial^2 v}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial v}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 v}{\partial \varphi^2} + \lambda v = 0, & \rho < b, 0 < \varphi < 2\pi \\ \text{边界条件: } v|_{\rho=b} = 0, & 0 < \varphi < 2\pi \end{cases}$$

令 $v(\rho, \varphi) = R(\rho)\Phi(\varphi)$

$$\text{则 } \frac{\rho^2}{R} \left(R'' + \frac{1}{\rho} R' + \lambda R \right) = -\frac{\Phi''}{\Phi} = \lambda_1 \quad \text{二次分离}$$

$$\begin{cases} \Phi''(\varphi) + \lambda_1 \Phi(\varphi) = 0 \\ \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - \lambda_1) R(\rho) = 0 \end{cases}$$

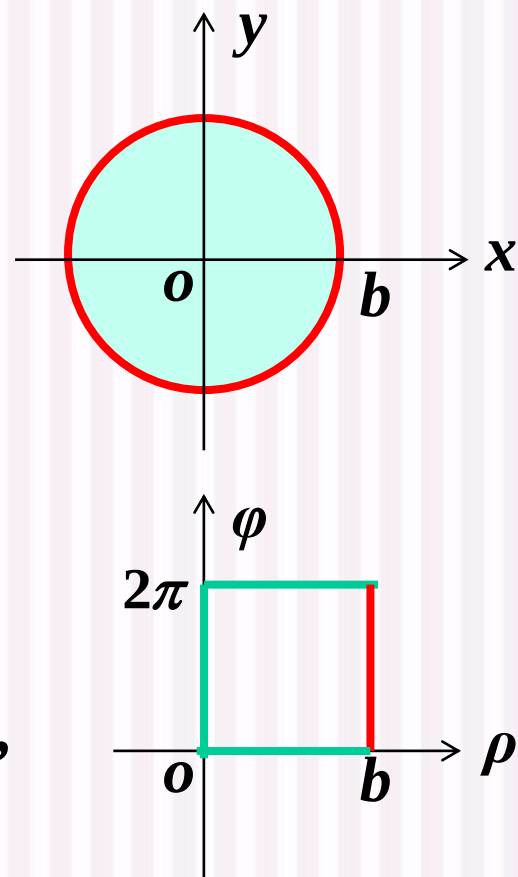
$$v|_{\rho=b} = 0 \Rightarrow R(b)\Phi(\varphi) = 0 \Rightarrow R(b) = 0 \quad \text{边界条件分离}$$

自然周期本征值问题:

$$\textcircled{1} \begin{cases} \Phi''(\varphi) + \lambda_1 \Phi(\varphi) = 0 \\ \Phi(0) = \Phi(2\pi) \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$$

本征值: $\lambda_1 = m^2, (m = 0, 1, 2, \dots)$

本征函数: $\Phi_m = A_m \cos m\varphi + B_m \sin m\varphi,$
 $(m = 0, 1, 2, \dots)$



Bessel方程的本征值问题:

$$\textcircled{2} \begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - m^2) R(\rho) = 0 \\ \text{边界条件: } R(b) = 0, |R(0)| < +\infty \end{cases}$$

Bessel方程的本征值问题

$$\begin{cases} \rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - m^2) R(\rho) = 0 \\ \text{边界条件: } R(b) = 0, |R(0)| < +\infty \end{cases}$$

(i) 当 $\lambda = 0$ 时, 方程变为Euler方程

× 无非零解

$$\rho^2 R''(\rho) + \rho R'(\rho) - m^2 R(\rho) = 0 \quad (m = 0, 1, 2, \dots)$$

$$\text{其通解为} \begin{cases} R_0(\rho) = C_0 + D_0 \ln \rho \quad (m = 0), \\ R_m(\rho) = C_m \rho^m + D_m \frac{1}{\rho^m} \quad (m > 0) \end{cases}$$

$$\text{由边界条件有} \begin{cases} |R(0)| < +\infty \Rightarrow D_m = 0 \quad (m = 0, 1, 2, \dots) \\ R(b) = 0 \Rightarrow C_m = 0 \quad (m = 0, 1, 2, \dots) \end{cases}$$

(ii) 当 $\lambda > 0$ 时,

$$\rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - m^2) R(\rho) = 0 \quad (1)$$

作代换 $r = \sqrt{\lambda} \rho$, 则 $\frac{dR}{d\rho} = \frac{dR}{dr} \frac{dr}{d\rho} = \sqrt{\lambda} \frac{dR}{dr}$, $\frac{d^2 R}{d\rho^2} = \lambda \frac{d^2 R}{dr^2}$

则方程(1)可转换为下列**标准 m 阶Bessel方程**:

$$r^2 R''(r) + r R'(r) + (r^2 - m^2) R(r) = 0$$

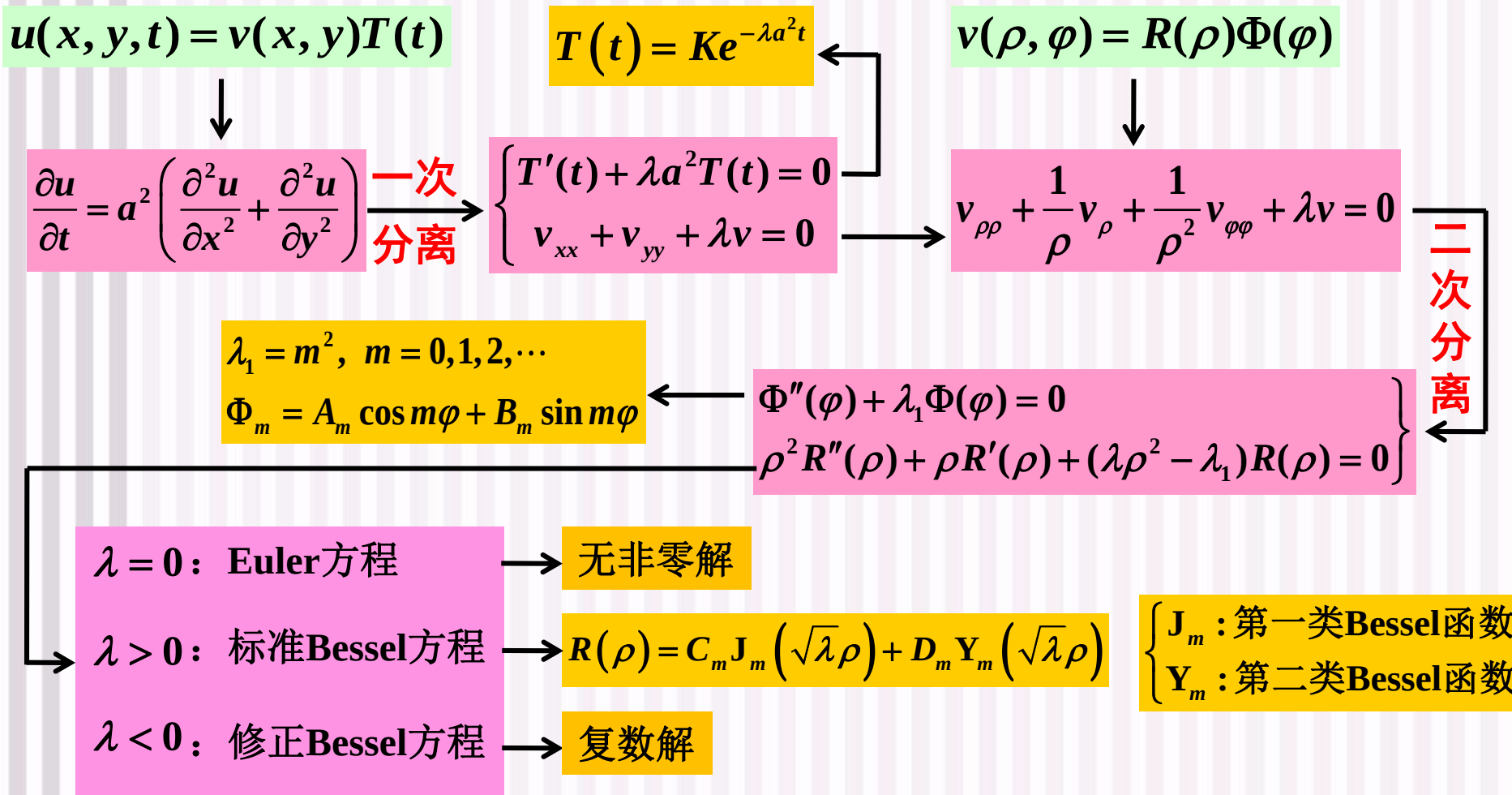
依据标准方程的通解(见5.2节)得方程(1)的通解为

$$R(\rho) = C_m J_m(\sqrt{\lambda} \rho) + D_m Y_m(\sqrt{\lambda} \rho), \quad m = 0, 1, 2, \dots$$

(iii) 当 $\lambda < 0$ 时, 修正Bessel方程

其解为复数解, 不做讨论.

三变量分离：两个本征方程





5.2 贝塞尔方程的求解

贝塞尔方程

◆ $x^2 y'' + xy' + (x^2 - \mu^2)y = 0$ —— 特殊函数方程

其中参数 μ 称为贝塞尔方程的阶数，
此时方程称为 μ 阶贝塞尔方程.

◆ $x^2 y'' + xy' + (\lambda x^2 - \mu^2)y = 0$ —— S-L本征方程

写为标准型：
$$\frac{d}{dx} \left[x \frac{dy}{dx} \right] - \frac{\mu^2}{x} y + \lambda xy = 0$$

其中 λ 为本征值参数， μ 是阶数.

贝塞尔方程的解

$$x^2 y'' + xy' + (x^2 - \mu^2)y = 0$$

阶数 μ 为任意实数或复数，仅讨论 $\mu \geq 0$ 的实数情形。

假定方程有如下形式的级数解：

$$y(x) = x^c (a_0 + a_1 x + \cdots + a_k x^k + \cdots) = \sum_{k=0}^{\infty} a_k x^{c+k} \quad (a_0 \neq 0)$$

$$xy'(x) = x \sum_{k=0}^{\infty} (c+k) a_k x^{c+k-1} = \sum_{k=0}^{\infty} (c+k) a_k x^{c+k}$$

$$x^2 y''(x) = \sum_{k=0}^{\infty} (c+k)(c+k-1) a_k x^{c+k}$$

$$\sum_{k=0}^{\infty} \left\{ [(c+k)(c+k-1) + (c+k) + (x^2 - \mu^2)] a_k x^{c+k} \right\} = 0$$

——5.2 贝塞尔方程的求解——

$$\text{化简得: } (c^2 - \mu^2)a_0x^c + \left[(c+1)^2 - \mu^2\right]a_1x^{c+1} \\ + \sum_{k=2}^{\infty} \left\{ \left[(c+k)^2 - \mu^2\right]a_k + a_{k-2} \right\} x^{c+k} = 0$$

$$(c^2 - \mu^2)a_0 = 0 \Rightarrow c = \pm \mu$$

$$\left[(c+1)^2 - \mu^2\right]a_1 = 0 \Rightarrow a_1 = 0, a_3 = a_5 \cdots = a_{2m-1} = 0$$

k 为奇数

$$\left[(c+k)^2 - \mu^2\right]a_k + a_{k-2} = 0 \Rightarrow a_k = -\frac{a_{k-2}}{k(2\mu+k)}$$

k 为偶数

$$a_2 = -\frac{a_0}{2(2\mu+2)}, \cdots, a_{2m} = (-1)^m \frac{a_0}{2^{2m} m! (\mu+1)(\mu+2) \cdots (\mu+m)}$$

(i) 当 $c = \mu$ 时, 可得一个特解

$$\begin{aligned} y(x) &= \sum_{k=0}^{\infty} a_k x^{c+k} = \sum_{m=0}^{\infty} a_{2m} x^{\mu+2m} \\ &= \sum_{m=0}^{\infty} (-1)^m \frac{a_0 x^{\mu+2m}}{2^{2m} m! (\mu+1)(\mu+2) \cdots (\mu+m)} \end{aligned}$$

Γ函数: $\Gamma(z) = \int_0^{+\infty} e^{-t} t^{z-1} dt \quad (\operatorname{Re}(z) > 0), \quad \Gamma(1) = 1, \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

$$\Gamma(z+1) = z\Gamma(z), \quad \frac{1}{\Gamma(-m)} = 0 \quad (m = 0, 1, 2, \dots)$$

$$\Gamma(\mu+m+1) = (\mu+m)(\mu+m-1) \cdots (\mu+2)(\mu+1)\Gamma(\mu+1)$$

$$\Gamma(\mu+m+1) = (\mu+m)! \quad (\text{当 } \mu \text{ 为 } 0 \text{ 或正整数时})$$

取 $a_0 = \frac{1}{2^\mu \Gamma(\mu+1)}$, 则特解

$$\begin{aligned} y(x) &= \sum_{m=0}^{\infty} (-1)^m \frac{a_0 x^{\mu+2m}}{2^{2m} m! (\mu+1)(\mu+2)\cdots(\mu+m)} \\ &= \sum_{m=0}^{\infty} (-1)^m \frac{x^{\mu+2m}}{2^{\mu+2m} m! (\mu+1)(\mu+2)\cdots(\mu+m) \Gamma(\mu+1)} \\ &= \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(\mu+m+1)} \left(\frac{x}{2}\right)^{\mu+2m} \\ &= J_\mu(x) \end{aligned}$$

称 $J_\mu(x)$ 为 μ 阶第一类贝塞尔函数

(ii) 当 $c = -\mu$ 时, 取 $a_0 = \frac{1}{2^{-\mu}\Gamma(-\mu+1)}$

可得另一特解

$$J_{-\mu}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(-\mu+m+1)} \left(\frac{x}{2}\right)^{-\mu+2m}$$

◆ 当 μ 不为整数时, 两个特解线性无关

此时可得通解: $y = AJ_{\mu}(x) + BJ_{-\mu}(x)$

特例: 当 μ 为半整数时, 解为初等函数.

◆ 当 μ 为整数时, $J_{-\mu}(x) = (-1)^{\mu} J_{\mu}(x)$, 两个特解线性相关

特例：当 μ 为半整数 $N + \frac{1}{2}$ (N 为自然数) 时，

贝塞尔方程的解为初等函数.

如： $\mu = \frac{1}{2}$ 时， $x^2 y'' + xy' + \left[x^2 - \left(\frac{1}{2} \right)^2 \right] y = 0$

作变量代换 $y(x) = \left(\frac{2}{\pi x} \right)^{\frac{1}{2}} u(x) \rightarrow u''(x) + u(x) = 0$

$$\therefore y(x) = A \left(\frac{2}{\pi x} \right)^{\frac{1}{2}} \sin x + B \left(\frac{2}{\pi x} \right)^{\frac{1}{2}} \cos x = AJ_{\frac{1}{2}}(x) + BJ_{-\frac{1}{2}}(x)$$

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x, \quad J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

半整数阶第一类贝塞尔函数为初等函数

证明当 μ 为整数时, $J_{-\mu}(x) = (-1)^\mu J_\mu(x)$

不妨设 $\mu = N$ (正整数),

$$\frac{1}{\Gamma(-K)} = 0 (k = 0, 1, \dots) \Rightarrow \frac{1}{\Gamma(-N + m + 1)} = 0 \quad (m = 0, 1, \dots, N - 1)$$

$$J_{-N}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(-N + m + 1)} \left(\frac{x}{2}\right)^{-N+2m}$$

$$= \sum_{m=N}^{\infty} \frac{(-1)^m x^{-N+2m}}{2^{-N+2m} m! \Gamma(-N + m + 1)}$$

$$= (-1)^N \left[\frac{x^N}{2^N N!} - \frac{x^{N+2}}{2^{N+2} (N+1)!} + \frac{x^{N+4}}{2^{N+4} (N+2)! 2!} - \dots \right]$$

$$= (-1)^N J_N(x)$$

对 μ 不为整数时的通解: $y = AJ_{\mu}(x) + BJ_{-\mu}(x)$

$$\text{取 } A = \cot \mu\pi, B = -\frac{1}{\sin \mu\pi} \quad (\mu \neq 1, 2, \dots)$$

$$\text{当 } \mu \text{ 为整数时, 令 } Y_{\mu}(x) = \lim_{\alpha \rightarrow \mu} \frac{J_{\alpha}(x) \cos \alpha\pi - J_{-\alpha}(x)}{\sin \alpha\pi}$$

则 $Y_{\mu}(x)$ 与 $J_{\mu}(x)$ 线性无关

称 $Y_{\mu}(x)$ 为 μ 阶第二类贝塞尔函数或诺依曼函数

贝塞尔方程的解为: $y(x) = AJ_{\mu}(x) + BY_{\mu}(x)$

其中 A, B 为任意常数, μ 为任意实数

总结：贝塞尔方程的解

$$x^2 y'' + xy' + (x^2 - \mu^2)y = 0 \quad (\text{阶数 } \mu \geq 0)$$

(i) 当 μ =半整数, 即 $\frac{1}{2}, \frac{3}{2}, \dots, k + \frac{1}{2} \dots$ (k 为自然数)时,

贝塞尔方程的解为初等函数.

(ii) 当 $\mu \neq$ 半整数时: $y(x) = AJ_\mu(x) + BY_\mu(x)$

其中 $J_\mu(x) = \sum_{m=0}^{+\infty} \frac{(-1)^m}{m! \Gamma(\mu + m + 1)} \left(\frac{x}{2}\right)^{\mu+2m} \rightarrow$ 第一类Bessel函数

$Y_\mu(x) = \begin{cases} \frac{J_\mu(x) \cos \mu\pi - J_{-\mu}(x)}{\sin \mu\pi}, & \mu \neq \text{整数} \\ \lim_{\alpha \rightarrow \mu} \frac{J_\alpha(x) \cos \alpha\pi - J_{-\alpha}(x)}{\sin \alpha\pi}, & \mu = \text{整数} \end{cases} \rightarrow$ 第二类Bessel函数



5.3 贝塞尔函数的性质

5.3.1 贝塞尔函数的基本形态及本征值问题

5.3.2 贝塞尔函数的递推公式

5.3.3 贝塞尔函数的正交性和模方

— Bessel函数的基本形态

第一类m阶Bessel函数 (考虑 $m = 0, 1, 2, \dots$)

$$J_m(x) = \sum_{k=0}^{+\infty} (-1)^k \frac{1}{k!(m+k)!} \left(\frac{x}{2}\right)^{m+2k}$$

如: $m = 0, J_0(x) = 1 - \left(\frac{x}{2}\right)^2 + \frac{1}{(2!)^2} \left(\frac{x}{2}\right)^4 - \frac{1}{(3!)^2} \left(\frac{x}{2}\right)^6 + \dots$

$$m = 1, J_1(x) = \frac{x}{2} - \frac{1}{2!} \left(\frac{x}{2}\right)^3 + \frac{1}{2!3!} \left(\frac{x}{2}\right)^5 - \frac{1}{3!4!} \left(\frac{x}{2}\right)^7 + \dots$$

.....

初值: $J_0(0) = 1, J_m(0) = 0 (m \neq 0)$

奇偶性: $J_m(-x) = (-1)^m J_m(x)$

第二类m阶Bessel函数

$$Y_0(x) = \frac{2}{\pi} J_0(x) \left(\ln \frac{x}{2} + C \right) - \frac{2}{\pi} \sum_{n=0}^{m-1} \frac{(-1)^n}{(n!)^2} \left(\frac{x}{2} \right)^{2n} \sum_{k=1}^n \frac{1}{k}$$

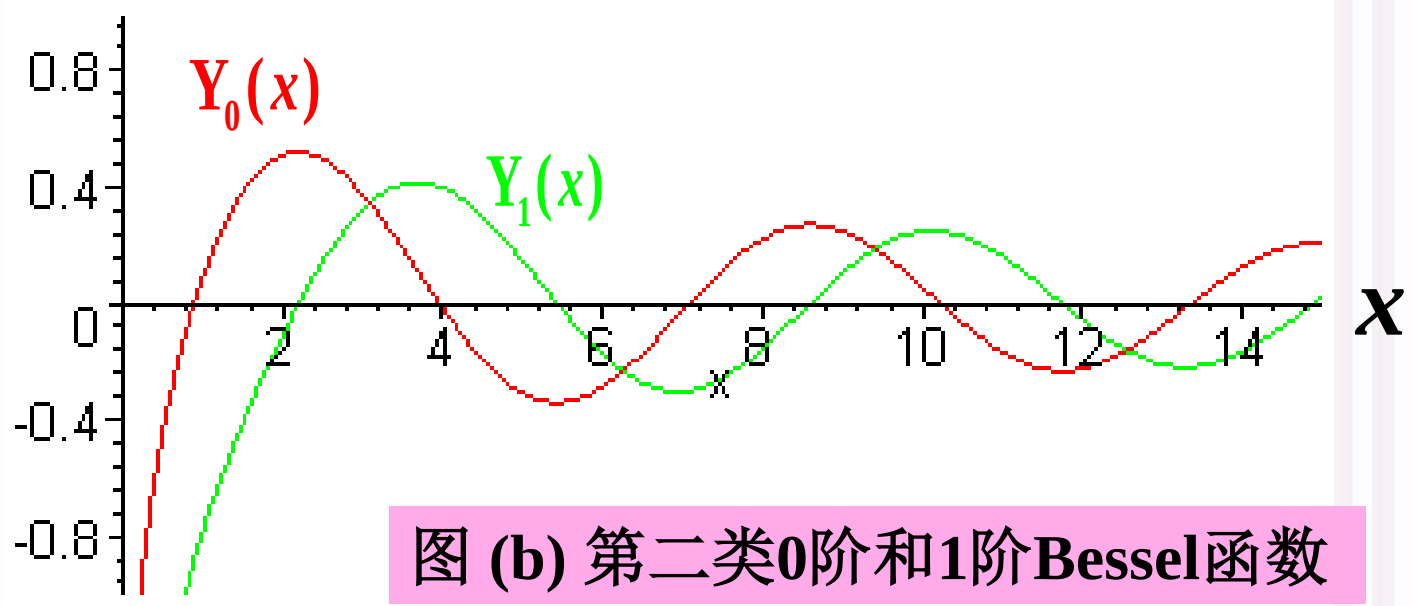
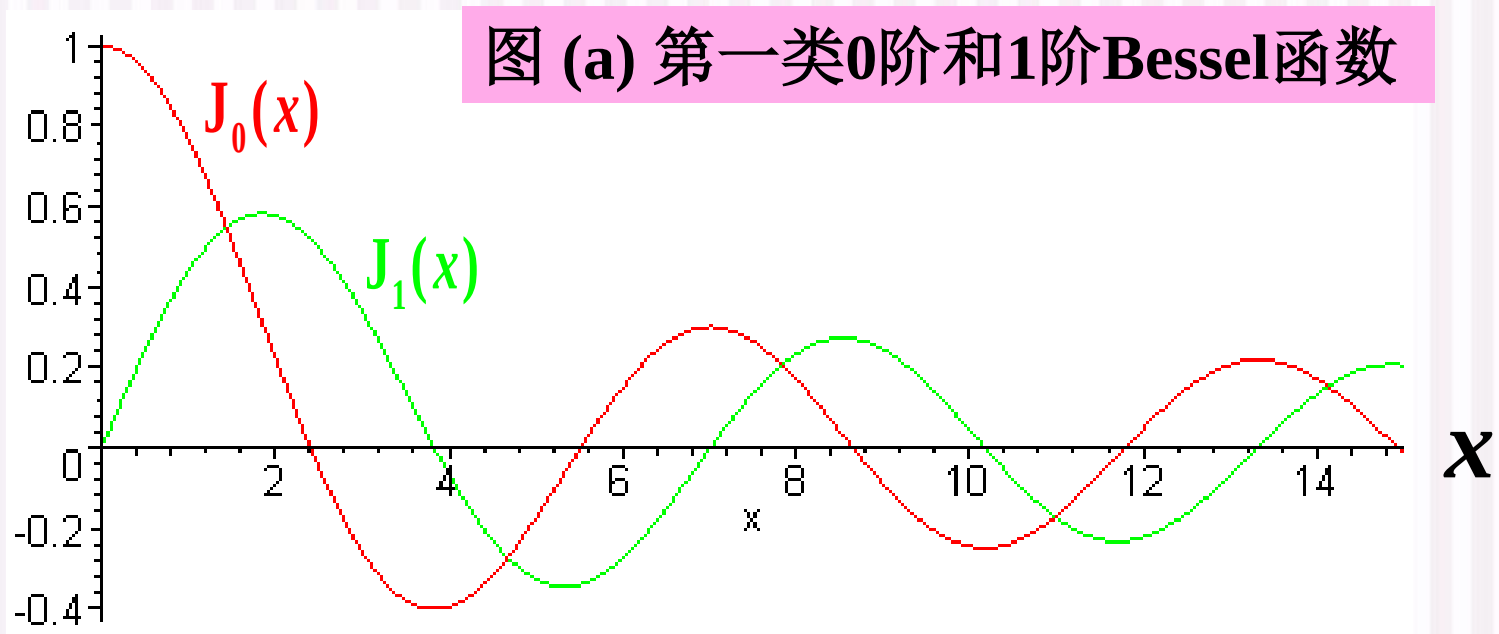
$$Y_m(x) = \lim_{\alpha \rightarrow m} \frac{J_\alpha(x) \cos \alpha\pi - J_{-\alpha}(x)}{\sin \alpha\pi} \quad (m = 1, 2, 3, \dots)$$

$$= \frac{2}{\pi} J_m(x) \left(\ln \frac{x}{2} + C \right) - \frac{1}{\pi} \sum_{n=0}^{m-1} \frac{(m-n-1)!}{n!} \left(\frac{x}{2} \right)^{-m+2n}$$

欧拉常数
 $C = 0.577$

$$- \frac{1}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(m+n)!} \left(\frac{x}{2} \right)^{m+2n} \left(\sum_{k=1}^{m+n} \frac{1}{k} + \sum_{k=1}^n \frac{1}{k} \right)$$

初值: $Y_m(0) = -\infty$ 奇偶性: $Y_m(-x) = (-1)^m Y_m(x)$



贝塞尔函数图像特点:

(1) 初值: $J_0(0)=1$, $J_m(0)=0$ ($m \neq 0$), $Y_m(0)=-\infty$;

贝塞尔方程 $\rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - m^2)R(\rho) = 0$ (1)

$\lambda > 0$ 时, $R(\rho) = C_m J_m(\sqrt{\lambda}\rho) + D_m Y_m(\sqrt{\lambda}\rho)$

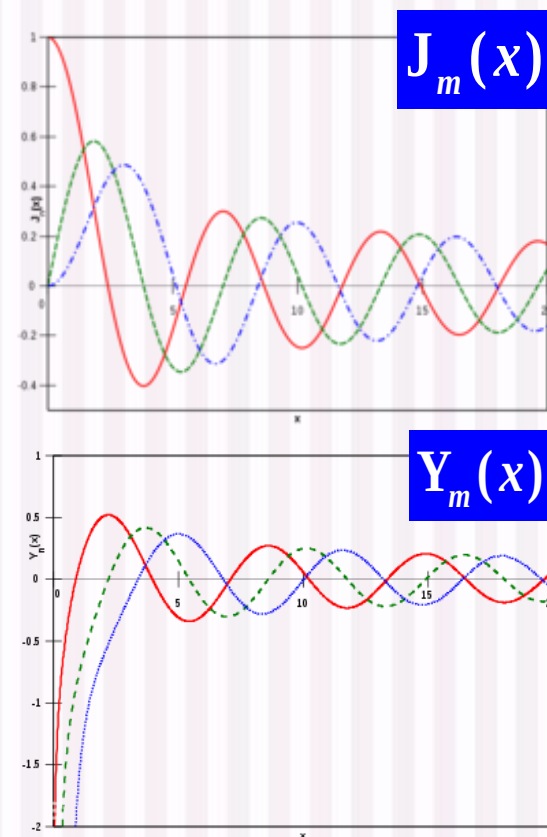
考虑边界条件: $R(b) = 0$, $|R(0)| < +\infty$

$$\Rightarrow D_m = 0 \Rightarrow C_m \neq 0$$

$\therefore \lambda > 0$ 时, 贝塞尔方程 (1) 的解为

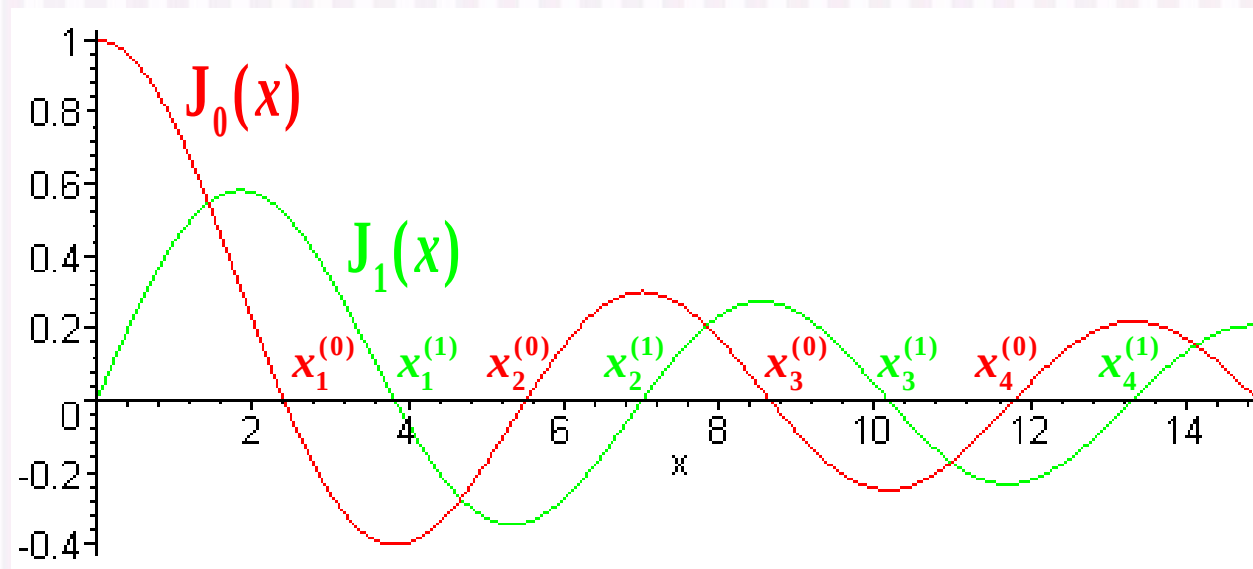
$$\begin{cases} R(\rho) = C_m J_m(\sqrt{\lambda}\rho), & 0 < \rho < b \\ R(b) = 0 \end{cases}$$

$$\Rightarrow J_m(\sqrt{\lambda}b) = 0 \Rightarrow \lambda = ?$$



(2) $J_m(x)$ 和 $Y_m(x)$ 都有无穷多个单重实零点, 且这些零点在 x 轴上关于原点对称分布;

(3) $J_m(x)$ 的零点与 $J_{m+1}(x)$ 的零点彼此相间分布;
若记 $x_n^{(m)}$ 表示 $J_m(x)$ 的第 n 个正零点,
则 $x_n^{(m)} < x_n^{(m+1)} < x_{n+1}^{(m)}$.



$J_0(x)$ 和 $J_1(x)$ 的前十个正零点

n	$x_n^{(0)}$	$x_n^{(1)}$
1	2.4048	3.8317
2	5.5201	7.0156
3	8.6537	10.1735
4	11.7925	13.3237
5	14.9309	16.4706
6	18.0711	19.6159
7	21.2116	22.7601
8	24.3526	25.9037
9	27.4935	29.0468
10	30.6346	32.1897

——5.3.1 贝塞尔函数的基本形态和本征值问题——

(4) $n \rightarrow \infty$ 时, $x_{n+1}^{(m)} - x_n^{(m)} \rightarrow \pi$, 即 $J_m(x)$ 的周期 $\rightarrow 2\pi$,

$$\text{且 } x \rightarrow \infty \text{ 时, } J_m(x) \Big|_{x \rightarrow \infty} \rightarrow \sqrt{\frac{2}{\pi x}} \cos \left(x - \frac{\pi m}{2} - \frac{\pi}{4} \right)$$

$$Y_m(x) \Big|_{x \rightarrow \infty} \rightarrow \sqrt{\frac{2}{\pi x}} \sin \left(x - \frac{\pi m}{2} - \frac{\pi}{4} \right)$$

根据贝塞尔函数零点的性质, 讨论 $\lambda > 0$ 时贝塞尔方程的解.

$$J_m(\sqrt{\lambda}b) = 0 \Rightarrow \sqrt{\lambda}b = x_n^{(m)}, n = 1, 2, \dots$$

$x_n^{(m)}$ 是 $J_m(x)$ 的第 n 个正零点

$$\therefore \begin{cases} \text{本征值: } \lambda_n = \frac{(x_n^{(m)})^2}{b^2} \quad (n = 1, 2, \dots), \\ \text{本征函数: } R_n(\rho) = C_m J_m \left(\frac{x_n^{(m)}}{b} \rho \right) \quad (n = 1, 2, \dots). \end{cases}$$



5.3 贝塞尔函数的性质

5.3.2 贝塞尔函数的递推公式

—— 5.3.2 贝塞尔函数的递推公式 ——

$$(1) \quad \frac{d}{dx} \left[\frac{J_{\mu}(x)}{x^{\mu}} \right] = - \frac{J_{\mu+1}(x)}{x^{\mu}}$$

$$J_{\mu}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(\mu + k + 1)} \left(\frac{x}{2} \right)^{\mu+2k}$$

证：左端 = $0 + \sum_{k=1}^{\infty} \frac{2k(-1)^k}{k! \Gamma(\mu + k + 1)} x^{2k-1} \left(\frac{1}{2} \right)^{\mu+2k}$

$$= \sum_{k=1}^{\infty} \frac{(-1)^k}{(k-1)! \Gamma(\mu + k + 1)} x^{2k-1} \left(\frac{1}{2} \right)^{\mu+2k-1}$$

$$= \sum_{l=0}^{\infty} \frac{-(-1)^l}{l! \Gamma(\mu + l + 2)} x^{2l+1} \left(\frac{1}{2} \right)^{\mu+2l+1}$$

$$= - \frac{1}{x^{\mu}} \sum_{l=0}^{\infty} \frac{(-1)^l}{l! \Gamma(\mu + 1 + l + 1)} \left(\frac{x}{2} \right)^{\mu+1+2l} = - \frac{J_{\mu+1}(x)}{x^{\mu}}$$

—— 5.3.2 贝塞尔函数的递推公式 ——

$$(2) \quad \frac{d}{dx} \left[x^\mu J_\mu(x) \right] = x^\mu J_{\mu-1}(x)$$

证：左端 = $\sum_{k=0}^{\infty} \frac{(2\mu+2k)(-1)^k}{k! \Gamma(\mu+k+1)} x^{2\mu+2k-1} \left(\frac{1}{2} \right)^{\mu+2k}$

$$= \sum_{k=0}^{\infty} \frac{2(\mu+k)(-1)^k}{k!(\mu+k)\Gamma(\mu+k)} x^\mu x^{\mu+2k-1} \left(\frac{1}{2} \right)^{\mu+2k}$$
$$= x^\mu \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(\mu-1+k+1)} \left(\frac{x}{2} \right)^{\mu-1+2k}$$
$$= x^\mu J_{\mu-1}(x)$$

$$J_\mu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(\mu+k+1)} \left(\frac{x}{2} \right)^{\mu+2k}, \quad \Gamma(\mu+k+1) = (\mu+k)\Gamma(\mu+k)$$

$$\mu \text{ 阶第一类Bessel函数 } J_{\mu}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(\mu + k + 1)} \left(\frac{x}{2} \right)^{\mu + 2k}$$

满足递推公式

$$\frac{d}{dx} \left[\frac{J_{\mu}(x)}{x^{\mu}} \right] = - \frac{J_{\mu+1}(x)}{x^{\mu}} \quad (1)$$

$$\frac{d}{dx} \left[x^{\mu} J_{\mu}(x) \right] = x^{\mu} J_{\mu-1}(x) \quad (2)$$

$$x J'_{\mu}(x) + \mu J_{\mu}(x) = x J_{\mu-1}(x) \quad (3)$$

$$x J'_{\mu}(x) - \mu J_{\mu}(x) = -x J_{\mu+1}(x) \quad (4)$$

$$\frac{2}{x} \mu J_{\mu}(x) = J_{\mu+1}(x) + J_{\mu-1}(x) \quad (5) \leftarrow (3) - (4)$$

$$2 J'_{\mu}(x) = J_{\mu-1}(x) - J_{\mu+1}(x) \quad (6) \leftarrow (3) + (4)$$

左端求出微商

递推公式的几点说明：

- 由公式(1)和(2)可得不定积分：

$$\frac{d}{dx} \left[\frac{J_{\mu}(x)}{x^{\mu}} \right] = -\frac{J_{\mu+1}(x)}{x^{\mu}} \quad (1)$$

两端积分 $\Rightarrow \int \frac{J_{\mu+1}(x)}{x^{\mu}} dx = -\frac{J_{\mu}(x)}{x^{\mu}} + C$

$$\frac{d}{dx} \left[x^{\mu} J_{\mu}(x) \right] = x^{\mu} J_{\mu-1}(x) \quad (2)$$

两端积分 $\Rightarrow \int x^{\mu} J_{\mu-1}(x) dx = x^{\mu} J_{\mu}(x) + C$

—— 5.3.2 贝塞尔函数的递推公式 ——

- 由公式 (3) $xJ'_\mu(x) - \mu J_\mu(x) = -xJ_{\mu+1}(x)$

有特例：当 $\mu = 0$ 时，

$$J_0'(x) = -J_1(x) \Leftrightarrow \int J_1(x) dx = -J_0(x) + C$$

- 由公式(5) $\frac{2}{x}\mu J_\mu(x) = J_{\mu+1}(x) + J_{\mu-1}(x)$ 有降阶公式，

$$J_{\mu+1}(x) = \frac{2\mu}{x} J_\mu(x) - J_{\mu-1}(x)$$

并可连续降阶至零阶和一阶表示。

- 第二类贝塞尔函数 $Y_\mu(x)$ 也有与第一类相同的递推公式。

$$J_{\mu+1}(x) = \frac{2\mu}{x} J_{\mu}(x) - J_{\mu-1}(x)$$

递推公式的应用：

$$\text{已知 } J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x, \quad J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

利用降阶公式计算高阶半整数阶Bessel函数

$$\begin{aligned} J_{\frac{3}{2}}(x) &= \frac{1}{x} J_{\frac{1}{2}}(x) - J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \left(-\cos x + \frac{1}{x} \sin x \right) \\ &= -\sqrt{\frac{2}{\pi}} x^{\frac{3}{2}} \cdot \frac{1}{x} \frac{d}{dx} \left(\frac{\sin x}{x} \right) = -\sqrt{\frac{2}{\pi}} x^{\frac{3}{2}} \cdot \left(\frac{1}{x} \frac{d}{dx} \right) \left(\frac{\sin x}{x} \right) \end{aligned}$$

同理可得

$$J_{-\frac{3}{2}}(x) = \sqrt{\frac{2}{\pi}} x^{\frac{3}{2}} \cdot \left(\frac{1}{x} \frac{d}{dx} \right) \left(\frac{\cos x}{x} \right)$$

一般地有

$$J_{n+\frac{1}{2}}(x) = (-1)^n \sqrt{\frac{2}{\pi}} x^{n+\frac{1}{2}} \cdot \left(\frac{1}{x} \frac{d}{dx} \right)^n \left(\frac{\sin x}{x} \right), n \text{ 为自然数}$$

$$J_{-(n+\frac{1}{2})}(x) = \sqrt{\frac{2}{\pi}} x^{n+\frac{1}{2}} \cdot \left(\frac{1}{x} \frac{d}{dx} \right)^n \left(\frac{\cos x}{x} \right), n \text{ 为自然数}$$

其中微分算子 $\left(\frac{1}{x} \frac{d}{dx} \right)^n$ 指算子 $\frac{1}{x} \frac{d}{dx}$ 连续作用 n 次

$$\text{如: } \left(\frac{1}{x} \frac{d}{dx} \right)^2 \left(\frac{\sin x}{x} \right) = \frac{1}{x} \frac{d}{dx} \left[\frac{1}{x} \frac{d}{dx} \left(\frac{\sin x}{x} \right) \right]$$

例：证明 (1) $\int x^n J_0(x) dx$

$$= x^n J_1(x) + (n-1)x^{n-1} J_0(x) - (n-1)^2 \int x^{n-2} J_0(x) dx;$$

证明：由 $\frac{d}{dx} [x^\mu J_\mu(x)] = x^\mu J_{\mu-1}(x)$, $J_0'(x) = -J_1(x)$ 有

$$\int x^n J_0(x) dx = \int x^{n-1} d[x J_1(x)]$$

$$= x^n J_1(x) - (n-1) \int x^{n-1} J_1(x) dx$$

$$= x^n J_1(x) + (n-1) \int x^{n-1} d(J_0(x))$$

$$= x^n J_1(x) + (n-1)x^{n-1} J_0(x) - (n-1)^2 \int x^{n-2} J_0(x) dx$$

—— 5.3.2 贝塞尔函数的递推公式 ——

$$(2) \int_0^a x^3 J_0\left(\frac{\mu}{a} x\right) dx = \frac{2a^4}{\mu^2} J_0(\mu), \text{ 其中 } \mu \text{ 是 } J_1(x) = 0 \text{ 的正根.}$$

证明： $(xJ_1(x))' = xJ_0(x), J_1(\mu) = 0$

$$\text{由 } \int x^n J_0(x) dx = x^n J_1(x) + (n-1)x^{n-1}J_0(x) - (n-1)^2 \int x^{n-2}J_0(x) dx$$

$$\text{有 } \int x^3 J_0(x) dx = x^3 J_1(x) + 2x^2 J_0(x) - 4x J_1(x) + C$$

用 $\frac{\mu}{a}x$ 替换 x , 并取 $[0, a]$ 上定积分

$$\begin{aligned} \text{则 } \int_0^a x^3 J_0\left(\frac{\mu}{a} x\right) dx &= \left[\frac{a}{\mu} x^3 J_1\left(\frac{\mu}{a} x\right) + 2 \frac{a^2}{\mu^2} x^2 J_0\left(\frac{\mu}{a} x\right) - 4 \frac{a^3}{\mu^3} x J_1\left(\frac{\mu}{a} x\right) \right] \Big|_0^a \\ &= \frac{a^4}{\mu} J_1(\mu) + \frac{2a^4}{\mu^2} J_0(\mu) - \frac{4a^4}{\mu^3} J_1(\mu) = \frac{2a^4}{\mu^2} J_0(\mu) \end{aligned}$$



5.3 贝塞尔函数的性质

5.3.3 贝塞尔函数的正交性和模方

一 Bessel函数的正交性

m 阶贝塞尔方程

$\rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - m^2)R(\rho) = 0$ 的本征解为:

本征值: $\lambda_n = \left(\frac{x_n^{(m)}}{b} \right)^2, (n = 1, 2, \dots)$, $x_n^{(m)}$ 是 $J_m(x)$ 的第 n 个正零点

本征函数: $R_n(\rho) = J_m(\sqrt{\lambda_n} \rho) = J_m\left(\frac{x_n^{(m)}}{b} \rho\right), (n = 1, 2, \dots)$.

记 $\lambda = \mu^2$, $J_m(\mu_k \rho)$ 和 $J_m(\mu_l \rho)$ 分别为本征值 μ_k^2, μ_l^2 对应的本征函数, 则可证

$$\int_0^b J_m(\mu_k \rho) J_m(\mu_l \rho) \rho d\rho = 0 \quad (k \neq l)$$

贝塞尔函数的加权正交性

证明：因为 $R_k(\rho) = J_m(\mu_k \rho)$, $R_l(\rho) = J_m(\mu_l \rho)$ 分别满足

$$\frac{d}{d\rho} \left[\rho \frac{dR_k(\rho)}{d\rho} \right] - \frac{m^2}{\rho} R_k(\rho) + \mu_k^2 \rho R_k(\rho) = 0 \quad \times R_l(\rho)$$

$$\frac{d}{d\rho} \left[\rho \frac{dR_l(\rho)}{d\rho} \right] - \frac{m^2}{\rho} R_l(\rho) + \mu_l^2 \rho R_l(\rho) = 0 \quad \times R_k(\rho)$$

相减并在 $[0, b]$ 上对 ρ 积分，得

$$\begin{aligned} \int_0^b R_l(\rho) \frac{d}{d\rho} \left[\rho \frac{dR_k(\rho)}{d\rho} \right] d\rho - \int_0^b R_k(\rho) \frac{d}{d\rho} \left[\rho \frac{dR_l(\rho)}{d\rho} \right] d\rho \\ + (\mu_k^2 - \mu_l^2) \int_0^b R_l(\rho) R_k(\rho) \rho d\rho = 0 \end{aligned}$$

$$\begin{aligned} \Leftrightarrow \int_0^b R_l(\rho) d \left[\rho \frac{dR_k(\rho)}{d\rho} \right] - \int_0^b R_k(\rho) d \left[\rho \frac{dR_l(\rho)}{d\rho} \right] \\ + (\mu_k^2 - \mu_l^2) \int_0^b R_l(\rho) R_k(\rho) \rho d\rho = 0 \end{aligned}$$

—— 5.3.3 贝塞尔函数的正交性和模方 ——

$$\begin{aligned} \int_0^b R_l(\rho) d\left[\rho \frac{dR_k(\rho)}{d\rho}\right] - \int_0^b R_k(\rho) d\left[\rho \frac{dR_l(\rho)}{d\rho}\right] &= R_l(\rho) \rho \frac{dR_k(\rho)}{d\rho} \Big|_0^b \\ &\quad - \int_0^b \rho \frac{dR_k(\rho)}{d\rho} \frac{dR_l(\rho)}{d\rho} d\rho - R_k(\rho) \rho \frac{dR_l(\rho)}{d\rho} \Big|_0^b + \int_0^b \rho \frac{dR_l(\rho)}{d\rho} \frac{dR_k(\rho)}{d\rho} d\rho \\ &= b \left[R_l(b) R_k'(b) - R_k(b) R_l'(b) \right] \end{aligned}$$

$$\because R_l(b) = J_m\left(\frac{x_l^{(m)}}{b} b\right) = 0, \quad R_k(b) = J_m\left(\frac{x_k^{(m)}}{b} b\right) = 0$$

$$\begin{aligned} \therefore \text{当 } k \neq l \text{ 时, } \int_0^b R_l(\rho) R_k(\rho) \rho d\rho &= - \frac{b \left[R_l(b) R_k'(b) - R_k(b) R_l'(b) \right]}{\mu_k^2 - \mu_l^2} = 0 \end{aligned}$$

$$\text{即 } \int_0^b J_m(\mu_k \rho) J_m(\mu_l \rho) \rho d\rho = 0 \quad (k \neq l)$$

二 Bessel函数的模方

$$\begin{aligned}\int_0^b J_m(\mu_k \rho) J_m(\mu_l \rho) \rho d\rho &= -\frac{b \left[R_l(b) R_k'(b) - R_k(b) R_l'(b) \right]}{\mu_k^2 - \mu_l^2} \\ &= -\frac{b \left[J_m(\mu_l b) J_m'(\mu_k b) \mu_k - J_m(\mu_k b) J_m'(\mu_l b) \mu_l \right]}{\mu_k^2 - \mu_l^2}\end{aligned}$$

$$\text{当 } l \rightarrow k \text{ 时, } \int_0^b \rho J_m^2(\mu_k \rho) d\rho = \int_0^b \rho J_m^2\left(\frac{x_k^{(m)}}{b} \rho\right) d\rho$$

$$\begin{aligned}&= -\lim_{\mu_l \rightarrow \mu_k} \frac{b \mu_k J_m(\mu_l b) J_m'(\mu_k b)}{\mu_k^2 - \mu_l^2} \\ &= -\lim_{\mu_l \rightarrow \mu_k} \frac{b \mu_k \left[b J_m'(\mu_l b) J_m'(\mu_k b) \right]}{-2 \mu_l} = \frac{b^2}{2} \left[J_m'(x_k^{(m)}) \right]^2\end{aligned}$$

$$\text{推导 } J'_m(x_k^{(m)}) = -J_{m+1}(x_k^{(m)})$$

$$\begin{aligned}\because -x^{-n}J_{n+1}(x) &= \frac{d}{dx} [x^{-n}J_n(x)] \\ &= x^{-n}J'_n(x) - nx^{-n-1}J_n(x)\end{aligned}$$

$$\begin{aligned}\therefore -\left(x_k^{(m)}\right)^{-n}J_{m+1}\left(x_k^{(m)}\right) \\ = \left(x_k^{(m)}\right)^{-n}J'_m\left(x_k^{(m)}\right) - n\left(x_k^{(m)}\right)^{-n-1}J_m\left(x_k^{(m)}\right)\end{aligned}$$

$$\therefore J'_m(x_k^{(m)}) = -J_{m+1}(x_k^{(m)})$$

↓
0

推导 $J'_m(x_k^{(m)}) = J_{m-1}(x_k^{(m)})$

$$\because x^n J_{n-1}(x) = \frac{d}{dx} [x^n J_n(x)] = x^n J'_n(x) + nx^{n-1} J_n(x)$$

$$\begin{aligned} \therefore \left(x_k^{(m)}\right)^n J_{m-1}\left(x_k^{(m)}\right) &= \left(x_k^{(m)}\right)^n J'_m\left(x_k^{(m)}\right) + n\left(x_k^{(m)}\right)^{n-1} J_m\left(x_k^{(m)}\right) \end{aligned}$$

0
↑

$$\therefore J_{m-1}\left(x_k^{(m)}\right) = J'_m\left(x_k^{(m)}\right)$$

$$\begin{aligned} \therefore \int_0^b \rho J_m^2(\mu_k \rho) d\rho &= \int_0^b \rho J_m^2\left(\frac{x_k^{(m)}}{b} \rho\right) d\rho && \text{模方公式} \\ &= \frac{b^2}{2} [J'_m(x_k^{(m)})]^2 = \frac{b^2}{2} [J_{m-1}(x_k^{(m)})]^2 = \frac{b^2}{2} [J_{m+1}(x_k^{(m)})]^2 \end{aligned}$$



5.6 贝塞尔函数在定解问题中的应用

三变量分离

$$u(\rho, \varphi, t) = R(\rho)\Phi(\varphi)T(t)$$



$$\begin{cases} \frac{\partial u}{\partial t} = a^2 \left(\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} \right) \\ u|_{\rho=b} = 0, \quad |u|_{\rho=0} < +\infty, \\ u|_{t=0} = f(\rho, \varphi). \end{cases}$$

分离
变量

$$\Phi''(\varphi) + \lambda_1 \Phi(\varphi) = 0$$

$$\begin{cases} \Phi(0) = \Phi(2\pi) \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$$

$$T'(t) + \lambda a^2 T(t) = 0$$

$$T(t) = A e^{-\lambda a^2 t}$$

$$\rho^2 R''(\rho) + \rho R'(\rho) + (\lambda \rho^2 - \lambda_1) R(\rho) = 0$$

$$\begin{cases} \lambda = \left(x_n^{(m)} / b \right)^2, \quad n = 1, 2, \dots \\ R_{mn}(\rho) = C_{mn} J_m \left(\rho x_n^{(m)} / b \right) \end{cases}$$

$$\begin{cases} R(b) = 0 \\ |R(0)| < +\infty \end{cases}$$

组合叠加得问题一般解：

$$u(\rho, \varphi, t) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \left(A_{mn} \cos m\varphi + B_{mn} \sin m\varphi \right) J_m \left(\frac{x_n^{(m)}}{b} \rho \right) e^{-\left(\frac{x_n^{(m)} a}{b} \right)^2 t}$$

其中 A_{mn} 和 B_{mn} 由初始条件 $u|_{t=0} = f(\rho, \varphi)$ 确定。

——5.6 贝塞尔函数在定解问题中的应用——

由初始条件 $u|_{t=0} = f(\rho, \varphi)$ 有

$$f(\rho, \varphi) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} (A_{mn} \cos m\varphi + B_{mn} \sin m\varphi) J_m \left(\frac{x_n^{(m)}}{b} \rho \right)$$

需将 $f(\rho, \varphi)$ 按 ρ 和 φ 进行双重展开, 即按 φ 进行傅里叶级数展开, 按 ρ 进行傅里叶-贝塞尔展开.

傅立叶—贝塞尔级数 (条件略):

$$f(\rho) \sim \sum_{n=1}^{\infty} C_n J_m \left(\frac{\mu_n^{(m)}}{b} \rho \right), \quad C_n = \frac{\int_0^b \rho f(\rho) J_m \left(\frac{\mu_n^{(m)}}{b} \rho \right) d\rho}{\frac{b^2}{2} J_{m+1}^2(\mu_n^{(m)})}$$

极坐标系下的轴对称问题

与角度 φ 无关的一类问题称为轴对称问题。

例 设有半径为1的薄均匀圆盘，其侧面绝缘，边界上的温度始终保持为零度，初始圆盘内温度分布为 $1-r^2$ ，其中 r 为圆盘内任一点的极半径，求圆盘的温度分布规律。

分析：由于是在圆域内求解问题，故采用极坐标。考虑到定解条件和 θ 无关，所以温度 u 只能是 r 和 t 的函数。

$$\left. \begin{aligned} \frac{\partial u}{\partial t} &= a^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \\ \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \varphi^2} \end{aligned} \right\} \xRightarrow{\text{轴对称}} \frac{\partial u}{\partial t} = a^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right)$$

解: 问题可归结为求下列定解问题, 设 $u = u(r, t)$

$$\begin{cases} \frac{\partial u}{\partial t} = a^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right), & 0 \leq r < 1 \\ u|_{t=0} = 1 - r^2 \\ u|_{r=1} = 0 \end{cases}$$

由物理意义有 $\begin{cases} |u| < \infty \\ u|_{t \rightarrow +\infty} = 0 \end{cases}$

令 $u(r, t) = F(r)T(t)$ 代入方程得

$$\frac{T'}{a^2 T} = \frac{F'' + \frac{1}{r} F'}{F} = -\lambda$$

$$T' + a^2 \lambda T = 0 \rightarrow T(t) = Ce^{-a^2 \lambda t}$$

$$u|_{t \rightarrow +\infty} = 0 \rightarrow \lambda > 0, \text{ 令 } \lambda = \beta^2, \text{ 则}$$

$r^2 F'' + rF' + \lambda r^2 F = 0$ 为零阶非标准贝塞尔方程,

通解为:

$$x^2 y'' + xy' + (x^2 - \mu^2)y = 0$$

$$F(r) = C_1 J_0(\beta r) + C_2 Y_0(\beta r)$$

$$\text{通解 } F(r) = C_1 J_0(\beta r) + C_2 Y_0(\beta r)$$

$$|u| < \infty \rightarrow C_2 = 0$$

$$u|_{r=1} = 0 \rightarrow J_0(\beta) = 0 \rightarrow \beta \text{ 是 } J_0(x) \text{ 的零点.}$$

用 $\mu_n^{(0)}$ ($n = 1, 2, \dots$) 表示 $J_0(x)$ 的正零点, 则

$$u_n(r, t) = T_n(t) F_n(r) = C_n e^{-a^2 (\mu_n^{(0)})^2 t} J_0(\mu_n^{(0)} r)$$

叠加得原问题的解为

$$u(r, t) = \sum_{n=1}^{\infty} C_n e^{-a^2 (\mu_n^{(0)})^2 t} J_0(\mu_n^{(0)} r)$$

由初始条件 $u|_{t=0} = 1 - r^2$ 得

$$1 - r^2 = \sum_{n=1}^{\infty} C_n J_0(\mu_n^{(0)} r)$$

傅里叶-贝塞尔级数
(广义傅里叶级数)

$$\begin{aligned} \text{故 } C_n &= \frac{2}{J_1^2(\mu_n^{(0)})} \int_0^1 (1 - r^2) J_0(\mu_n^{(0)} r) r \, dr \\ &= \frac{2}{J_1^2(\mu_n^{(0)})} \left[\int_0^1 r J_0(\mu_n^{(0)} r) \, dr - \int_0^1 r^3 J_0(\mu_n^{(0)} r) \, dr \right] \end{aligned}$$

由递推公式 $\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$

有： $d[(\mu_n^{(0)} r) J_1(\mu_n^{(0)} r)] = (\mu_n^{(0)} r) J_0(\mu_n^{(0)} r) d(\mu_n^{(0)} r)$

即： $d[r J_1(\mu_n^{(0)} r)] = \mu_n^{(0)} r J_0(\mu_n^{(0)} r) dr$

$$\therefore \int_0^1 r J_0(\mu_n^{(0)} r) dr = \frac{r J_1(\mu_n^{(0)} r)}{\mu_n^{(0)}} \bigg|_0^1 = \frac{J_1(\mu_n^{(0)})}{\mu_n^{(0)}}$$

$$\int_0^1 r^3 J_0(\mu_n^{(0)} r) dr = \int_0^1 r^2 d \left[\frac{r J_1(\mu_n^{(0)} r)}{\mu_n^{(0)}} \right] = \frac{J_1(\mu_n^{(0)})}{\mu_n^{(0)}} - \frac{2 J_2(\mu_n^{(0)})}{(\mu_n^{(0)})^2}$$

—— 5.6 贝塞尔函数在定解问题中的应用 ——

$$\begin{aligned} C_n &= \frac{2}{J_1^2(\mu_n^{(0)})} \left[\int_0^1 r J_0(\mu_n^{(0)} r) \, dr - \int_0^1 r^3 J_0(\mu_n^{(0)} r) \, dr \right] \\ &= \frac{2}{J_1^2(\mu_n^{(0)})} \left[\frac{2J_2(\mu_n^{(0)})}{(\mu_n^{(0)})^2} \right] = \frac{4J_2(\mu_n^{(0)})}{(\mu_n^{(0)})^2 J_1^2(\mu_n^{(0)})} \end{aligned}$$

$$\begin{aligned} \therefore u(r, t) &= \sum_{n=1}^{\infty} C_n e^{-a^2(\mu_n^{(0)})^2 t} J_0(\mu_n^{(0)} r) \\ &= \sum_{n=1}^{\infty} \frac{4J_2(\mu_n^{(0)})}{(\mu_n^{(0)})^2 J_1^2(\mu_n^{(0)})} J_0(\mu_n^{(0)} r) e^{-a^2(\mu_n^{(0)})^2 t} \end{aligned}$$

其中 $\mu_n^{(0)}$ 是 $J_0(r)$ 的正零点.

第5章 贝塞尔函数

- **熟练掌握：**
 - 1、第一类贝塞尔函数的递推公式，利用递推公式进行证明和积分；
- **掌握：**
 - 1、贝塞尔方程和贝塞尔函数的关系
 - 2、贝塞尔函数的零点性质；
 - 3、贝塞尔方程和贝塞尔函数的本征值问题；
- **理解：**
 - 1、贝塞尔函数的正交性；
 - 2、用贝塞尔函数求解定解问题

第5章 作业

教材P128-130: 6, 7, 17